

Lab-o: Familiarizing with Numpy, Pandas and Matplotlib

1. Create 1-D and 2-D arrays using Numpy. Also convert 1-D array to appropriate 2-D using reshape() and add two arrays. Also find the shape of a 2-D array using shape().
2. Use an appropriate indexing method to access the last data of the 2-D array.
3. For the above 1-D and 2-D array created, slice the arrays to get specific data you want. You can also use negative slicing
4. Search the desired data using where() and searchsorted()
5. Calculate mean, median, mode, standard deviation and variance for a 1-D data. Also find the MSE and RMSE
6. Create a Dataframe using dictionaries and numpy arrays.
7. Import datasets using read_csv() and visualize the data using head() and tail(). Use describe() and info() to find the description of the data.
8. Add new columns to the existing dataframe and add any two columns and store the result in the third column
9. Create a line plot, scatter plot, bar graph and pie chart with labels using matplotlib for a given series of data

Subjects	Marks
Probability and Statistics	60
Data Science and Machine Learning	45
Artificial Intelligence	59
Computer Graphics	72
Engineering Economics	80